The three-color Ramsey number of the 10-vertex path

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Abstract

Writing P_n for the path on n vertices and $R_3(P_n) = R(P_n, P_n, P_n)$ for its three-color Ramsey number, Gyárfás, Ruszinkó, Sárközy and Szemerédi proved $R_3(P_n) = 2n - 2 + (n \bmod 2)$ for all sufficiently large n [18, author manuscript, July 9, 2012, Theorem 1]. Dybizbański, Dzido and Radziszowski (DDR) established $R_3(P_8) = 14$ and $R_3(P_9) = 17$ [8, arXiv:1207.5851v3, Theorems 10 and 8 with Section 2], completing the cases $n \le 9$ recorded by Radziszowski's survey [31, revision 18, p. 65, Section 6.4.1(c)]. We prove the next case, $R_3(P_{10}) = 18$: every three-coloring of the edges of K_{18} contains a monochromatic ten-vertex path, while a coloring of K_{17} contains none. The new bound is the upper bound 18; the lower bound uses the even-case K_4 blow-up in [18, Section 1.1, after Theorem 1] and [8, Section 2]. This is the ten-vertex case of Conjecture 2 in the cited DDR version, which credits Faudree and Schelp's 1975 suggestion. Boolean satisfiability refutations prove the upper bound. Lean checks the full certificate proof using trusted native evaluation; the default Lake build alone is conditional.

The proofs, the Lean formalization, the search code, the certificate checkers and the text were produced by AI models (from OpenAI and Anthropic), working under the author's direction.

1 The result

For graphs G_1, G_2, G_3, the Ramsey number $R(G_1, G_2, G_3)$ is the least N such that every coloring of the edges of K_N with colors $1, 2, 3$ contains, for some i, a copy of G_i all of whose edges have color i. We write $R_3(G) = R(G, G, G)$, and P_n for the path on n vertices (with $n-1$ edges). Paths are not required to be induced.

Theorem 1.1. $R_3(P_{10}) = 18$. *That is, every coloring of the edges of K_{18} with three colors contains a monochromatic path on 10 vertices, and some 3-coloring of the edges of K_{17} contains none.*

The advance is the upper bound $R_3(P_{10}) \le 18$. Gyárfás, Ruszinkó, Sárközy and Szemerédi proved $R_3(P_n) = 2n - 2 + (n \bmod 2)$ for all sufficiently large n [18, author manuscript, July 9, 2012, Theorem 1], with its corrigendum. Dybizbański, Dzido and Radziszowski proved $R_3(P_8) = 14$ (Theorem 10) and $R_3(P_9) \le 17$ (Theorem 8), which with their Section 2 lower bound gives $R_3(P_9) = 17$ [8, arXiv:1207.5851v3]. Radziszowski's survey [31, revision 18, p. 65, Sections 6.4 and 6.4.1(c)] records the formula through $n = 9$ and P_{10} as the first open case. Our K_{17} lower bound is the even-case blow-up in [18, Section 1.1, after Theorem 1], also given in [8, arXiv:1207.5851v3, Section 2]. The latter attributes Conjecture 2's suggestion to Faudree and Schelp (1975), quoting "they feel"; its closing remarks say, "We did not complete these proofs", referring to P_{10} and P_{11}. This paper settles $n = 10$ of Conjecture A.1; the general conjecture remains open.

Heule, Kullmann, Wieringa and Biere's cube-and-conquer [20, author manuscript, Section 3, Overview] "partitions a formula into many subformulas". We use the verified LRAT checker from

the `bv_decide` work of Böving et al. [2, version of record, introduction, contributions, and Section 4.4], which lists an "LRAT UNSAT proof checker". Szeider's streaming tool `lrat-catcher` [34, arXiv:2607.00815v2, abstract] "turns a certificate into a Lean theorem" using Lean core's verified checker. Our certificate import uses the same checker, with its own compact certificate encoding (Appendix F.3).

For the verification method, Kirchweger, Manrique and Szeider [23, IJCAR 2026, abstract] describe "a unified, machine-checked verification pipeline" for graph generation.

Related computations include Heule, Kullmann and Marek's Boolean Pythagorean triples proof [19, author preprint, Theorem 1], Codish, Frank, Itzhakov and Miller's "$R(4,3,3) = 30$" [5, author preprint, abstract], and Heule and Scheucher's "An empty hexagon in every set of 30 points" [21, TACAS 2024, title], with Subercaseaux et al.'s "Formal verification of the empty hexagon number" [33, ITP 2024, title]. Wesley's Theorems 3 and 4 give $R(C_3, C_6, C_6) = R(C_5, C_6, C_6) = 15$ [36, arXiv:2509.03784v1]. Galindez reports "$R(C_{10}, C_{10}, C_{10}) \leq 26$" [16, Zenodo record title], using cube-and-conquer and Kirchweger–Szeider's "SAT modulo symmetries" [25, TOCL 2024, title]; Shao, Jiang, Xu and Wang provide the "reproduction package: $R(C_4, C_6, C_8) = 12$" [32, Zenodo record title]. Posted work also includes Leo-Y-Zhang's "CycleRamsey" [41, repository title] and Zhang's k-color tree-bound corollary $R_k(T) \leq k(n-2)+2$ for trees on $n \geq 2$ vertices [42, pinned README, Ramsey corollary], from the GPT-6 Astra proof packaged by Adamczewski; it gives $R_3(P_{10}) \leq 26$. Allen et al. consider "Ramsey-type problems for paths and cycles" [1, arXiv:2607.03243v1, title].

For "Ramsey numbers for paths", see Dzido [10, title] and Dzido–Fidytek [9, title]; Omidi–Raeisi treat "paths versus cycles" [30, title], and Maherani et al. give mixed-path cases [27, arXiv:1207.3771v1, Theorem 1.1]. Earlier cycle work includes Benevides–Skokan's "even cycles" [4, title], Figaj–Łuczak's "long even cycles" [14, title], and Letzter's "connected matchings method" [26, title]. Small-case credit to Irving, Yang–Rowlinson and Yang [22, 39, 40, 38] follows the survey [31, revision 18, p. 65, Section 6.4.1(c)]. The extremal comparison credits Erdős–Gallai's "maximal paths and circuits" [11, title] and the Faudree–Schelp theorem [15] as recalled in DDR [8, arXiv:1207.5851v3, Theorem 4]. Kopylov's "Maximal paths and cycles in a graph" [24, title] gives related context. We use de Moura and Ullrich's Lean 4 [29] and the mathlib Community's library [28]. The separate checks use Wetzler, Heule and Hunt's `drat-trim` [37, title] for conversion to Cruz-Filipe et al.'s LRAT format [7, abstract], and Tan, Heule and Myreen's `cake_lpr` [35, title] for checking.

2 Proof overview

The difficult step is obtaining a tractable case split without losing a coloring: fixing a guaranteed path and imposing symmetry constraints must be jointly sound, and discarded branches leave a coverage obligation. A lexicographic minimum and a separate cover refutation discharge these obligations. The earlier obstruction is explicit in the closing remarks of Dybizbański, Dzido and Radziszowski [8, arXiv:1207.5851v3, Section 5]: "the complexity of case analysis appears to be quite harder". The reduction starts with a monochromatic P_9, as in the bound of Dybizbański, Dzido and Radziszowski, Theorem 8 of the version cited above. Here that bound is re-derived by smaller Boolean satisfiability (SAT) refutations, starting with P_3 on K_5 and ending with P_9 on K_{17}. Each refutation fixes only the path guaranteed by the preceding one. The construction in Figure 1 supplies the matching lower bound; Appendix B gives its proof.

A Boolean literal is a variable or its negation; a clause is a disjunction of literals. Each logged path clause is rebuilt in Lean from its color and distinct vertices. Refuting this subset suffices:

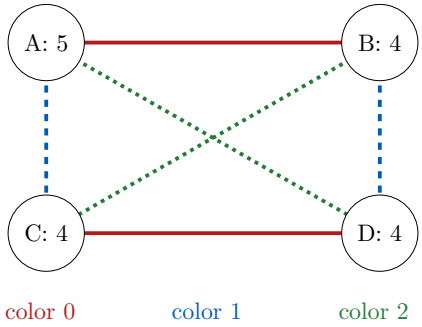


Figure 1: The K_{17} coloring, compressed into four parts. Each line represents all edges between its parts; all edges within a part have color 0. In each color, the two label classes have sizes 9 and 8.

a coloring without a monochromatic path would satisfy every path clause, hence also the logged subset. Cubes are conjunctions of literals fixing edge colors. A separate cover refutation proves that every model satisfies at least one listed cube; each cube has its own refutation. Lemma E.1 joins these facts. Appendices C–E give the encoding, the reductions and this complete dependency chain.

3 What is checked and trusted

In namespace `R3P10`, the unconditional theorem is `R3_P10_eq_18 : ramsey3Path 10 = 18`. Here `Col n` is `Sym2 (Fin n) → Fin 3`: colors on unordered vertex pairs. Diagonal values are unused and arbitrary; colors $0, 1, 2$ replace $1, 2, 3$. The predicate `HasMonoPath n m X` means that there exist a color and an injective map from m vertices to n vertices whose consecutive pairs have that color. It requires no condition on other pairs, so it means our non-induced P_m, with m vertices, not edges. The definition `ramsey3Path m` is the infimum in the natural numbers of the n for which every `Col n` satisfies that predicate. The proof establishes membership at 18 and a countercoloring at 17; restriction of that coloring excludes every smaller n. Thus this infimum is precisely the Ramsey number in Theorem 1.1, with no remaining certificate hypotheses.

Lean proves the lower bound, symmetry reductions, encoding soundness and assembly of refutations. Its verified LRAT checker checks a line-indexed format of propositional refutations. The final theorem's axioms are `propext`, `Classical.choice`, `Quot.sound`, `Lean.ofReduceBool` and `Lean.trustCompiler`. The last two support native evaluation of certificate checks, list lengths and the parsed-formula comparison. The proof trusts Lean's kernel, compiler and imported compiled modules; rebuilding the certificate modules avoids trusting previously built copies. The SAT solver, converter and external checkers are outside this trusted base. Literature attributions, historical timings and reports of separate checker executions are outside Lean. Appendix F gives the declarations and dependency details.

From the accompanying folder, `cd lean`, `lake exe cache get`, then `lake build` check the reduction, lower bound and conditional theorem, not the unconditional value. The full command `python3 checks/check.py full` regenerates and checks every certificate and builds `R3P10Final`. Installing the archived certificates [17] first avoids repeating the search. Quick mode checks a sample and cannot replace the full check. See `COMPLETION.md` for the completed full check and `checks/README.md` and `lean/README.md` for requirements, commands and independent checks. The proof is computer-assisted and has not been reviewed by human experts.

Appendix A The conjecture

The general conjecture is Conjecture 2 of Dybizbański, Dzido and Radziszowski [8, arXiv:1207.5851v3, Section 2]; its ten-vertex case is proved here.

Conjecture A.1 (suggested by Faudree–Schelp [15]; stated as a conjecture in [8]; see [31])**.** *For all* $n \geq 1$, $R_3(P_n) = 2n - 2 + (n \bmod 2)$.

Appendix B The lower bound

Partition $\{0, \ldots, 16\}$ into $A = \{0, \ldots, 4\}$, $B = \{5, \ldots, 8\}$, $C = \{9, \ldots, 12\}$ and $D = \{13, \ldots, 16\}$. Color an edge with color 0 if both ends lie in $A \cup B$ or both lie in $C \cup D$; with color 1 if it joins A to C or B to D; and with color 2 if it joins A to D or B to C. This is the blow-up of the factorization of K_4 into the matchings $\{AB, CD\}$, $\{AC, BD\}$, $\{AD, BC\}$, with the edges inside the parts given color 0. For this construction, Gyárfás, Ruszinkó, Sárközy and Szemerédi [18, author manuscript, July 9, 2012, Section 1.1, paragraph following Theorem 1] write: "One type comes from a 'blow-up' of a factorization of K_4." The odd and even part sizes are specified there; see also Dybizbański, Dzido and Radziszowski [8, arXiv:1207.5851v3, Section 2, paragraph following Conjecture 2].

Proposition B.1. *This coloring of* K_{17} *has no monochromatic* P_{10}*. Hence* $R_3(P_{10}) \geq 18$*.*

Proof. Each color has two label classes: $A \cup B$ and $C \cup D$ for color 0, $A \cup C$ and $B \cup D$ for color 1, $A \cup D$ and $B \cup C$ for color 2, of sizes 9 and 8 in each case. The edge invariant is that every edge of a color lies inside one of that color's two label classes; no edge of that color joins them. So every connected piece of a color class lies inside one label class and has at most 9 vertices, and a monochromatic path, being connected, has at most 9 vertices. □

In Lean the coloring is written out edge by edge, and the argument is a general lemma (for all n and m: if each color has a labeling of the vertices that is constant along edges of that color and whose classes have fewer than m vertices, there is no monochromatic P_m) together with two finite facts decided by the kernel's own reduction (`decide +kernel`), without native evaluation.

Appendix C The encoding

For the upper bound, suppose a coloring of K_{18} has no monochromatic P_{10}. Relabel vertices and colors so that $0, 1, \ldots, 8$ is a path of color 0. Among colorings reachable by permutations that preserve this path, take a lexicographically least one. It satisfies every imposed symmetry inequality simultaneously (Lemmas C.1 and C.2). The chosen symmetries need not include every permutation preserving the path.

A Boolean literal is a variable or its negation; a clause is a disjunction of literals. A model assigns truth values satisfying every clause; a refutation certifies that no such assignment exists. The SAT variables specify one color per edge, with auxiliary variables expressing the symmetry inequalities. A path clause excludes a specified monochromatic P_{10}. Rather than list all paths, the solver logs the path clauses it uses. Each is independently recognized as a path clause and rebuilt in Lean from its color and distinct vertices. Refuting this subset suffices: a coloring without a monochromatic path would satisfy every path clause, hence also the logged subset. The solver's path-finding procedure therefore need not be trusted for completeness.

The remaining formula is divided into cubes, conjunctions of literals fixing edge colors. A separate cover refutation proves that every model satisfies at least one listed cube; each cube has

its own refutation. Lemma E.1 joins these facts, contradicting the assumed coloring. The same argument establishes each smaller bound used to fix a path. Appendices C–E give the encoding, the reductions and this complete dependency chain.

Throughout, a *coloring* of K_n is a map X from the edges to $\{0,1,2\}$, and it is *good* (for m) if it has no monochromatic P_m. We order the edges $\{i,j\}$, $i<j$, row by row: $\{0,1\},\{0,2\},\dots,\{0,n-1\},\{1,2\},\dots,\{n-2,n-1\}$, and number them $e=0,1,\dots,\binom{n}{2}-1$ in this order.

C.1 Variables and the coloring clauses

The variable $x_{e,c}=3e+c+1$ means "edge e has color c". An unordered pair in the edge subscript denotes its number in the preceding edge order. For each edge there is one clause $x_{e,0}\vee x_{e,1}\vee x_{e,2}$ and three clauses $\neg x_{e,c}\vee\neg x_{e,d}$, $c<d$, so that models correspond to colorings. For a color c and distinct vertices $p_0,\dots,p_{m-1}$, the *path clause*

$$\bigvee_{i=0}^{m-2}\neg x_{\{p_i,p_{i+1}\},c}$$

says that this path is not monochromatic in color c. A coloring is good exactly when its assignment satisfies every path clause. The path clauses are not written into the formula in advance; they are generated during the search (Section C.4).

C.2 The fixed path

Vertex permutations $\pi\in S_n$ and color permutations $\sigma\in S_3$ act on colorings by $(g\cdot X)(\{\pi a,\pi b\})=\sigma(X(\{a,b\}))$ for $g=(\pi,\sigma)$. They map good colorings to good colorings.

Here "good" means having no monochromatic P_m, for the same m as in the preceding encoding. The following lemma fixes only path edges.

Lemma C.1 (fixed path)**.** *Let $k\le n$ and suppose every 3-coloring of K_n contains a monochromatic P_k. Then every good coloring of K_n is equivalent under $S_n\times S_3$ to a good coloring X with $X(\{t,t+1\})=0$ for $0\le t\le k-2$.*

Proof. Take a monochromatic path $v_0,\dots,v_{k-1}$ in color c; apply a vertex permutation sending v_t to t and the transposition of colors c and 0. □

The formula for the rung "P_m on K_n with a fixed P_k" contains the $k-1$ unit clauses $x_{\{t,t+1\},0}$. Only the path edges are constrained; the chords of the fixed path are free.

C.3 Lex-leader constraints

Compare colorings by the lexicographic order $\le_{\text{lex}}$ of their color vectors $(X(e))_e$ in edge order, with $0<1<2$. Let Γ be a list of generators $g=(\pi,\sigma)$. With a fixed P_k on $0,\dots,k-1$ and $W=\{k,\dots,n-1\}$, Γ consists of the transpositions of pairs of vertices of W (in lexicographic order), the reversal $t\mapsto k-1-t$ of $\{0,\dots,k-1\}$, and the color transposition $1\leftrightarrow 2$. On the first rung, K_5, nothing is fixed and Γ consists of all vertex transpositions and the color transpositions $0\leftrightarrow 1$ and $1\leftrightarrow 2$.

In the following lemma, goodness still refers to P_m and Γ is the preceding list of path-preserving vertex and color permutations.

Lemma C.2 (lex-leader)**.** *Let S be the set of good colorings with color 0 on the fixed path. Each $g \in \Gamma$ maps S to itself. Consequently, if S is nonempty it contains a coloring X^* with $X^* \leq_{\text{lex}} g \cdot X^*$ for every $g \in \Gamma$.*

Proof. When a path is fixed, each vertex permutation in Γ maps the path's edge set to itself and each color permutation in Γ fixes color 0, so Γ preserves the fixed path; every g preserves goodness. (With nothing fixed, S is the set of all good colorings.) Take X^* to be the $\leq_{\text{lex}}$-least coloring reachable from a member of S by finitely many moves from Γ; it lies in S, and $g \cdot X^*$ is reachable too, so $X^* \leq_{\text{lex}} g \cdot X^*$. □

The point of Lemma C.2 is that the lex-leader constraints for *every* generator hold simultaneously for one coloring that also has the fixed path; the two reductions are therefore jointly sound. The generators do not need to generate the full stabilizer of the fixed path: a weaker set of constraints only leaves more symmetry in the search.

The constraint $X \leq_{\text{lex}} g \cdot X$ is encoded in the standard way [6, technical report TR-1996-012, Section 3, Proposition 3.1 and the following auxiliary-variable clauses]. The clauses below adapt that Boolean-variable construction to three colors. The positions compared are the edges f in edge order, except those with $\pi^{-1}(f) = f$ and σ the identity (where $X(f) = (g \cdot X)(f)$ identically). For the j-th compared position f_j, let $Y_j = \sigma(X(\pi^{-1} f_j))$ be the color of $g \cdot X$ there, and let q_j be an auxiliary variable meaning "X and $g \cdot X$ agree on the first j compared positions" (q_0 is true and omitted). The clauses are, for colors a, b:

$$q_{j-1} \to \neg(X(f_j) = a \wedge Y_j = b) \quad (a > b), \qquad q_{j-1} \wedge X(f_j) = a \wedge Y_j = a \to q_j,$$
$$q_j \to q_{j-1}, \qquad q_j \wedge X(f_j) = a \to Y_j = a.$$

Under the intended meaning of the q_j, every clause holds exactly when $X \leq_{\text{lex}} g \cdot X$. For K_{18} with a fixed P_9 there are 38 generators (36 transpositions of the nine vertices of W, the reversal, and the color swap), and the base formula has 1,868 variables (459 color variables and 1,409 auxiliaries) and 14,672 clauses.

C.4 Path clauses generated during the search

The solver is CaDiCaL [3] with an external propagator attached through the IPASIR-UP interface [12, 13]. The propagator follows the assignment of the color variables. Whenever an edge receives a color c, it searches for paths on m vertices through that edge, in the graph of edges not yet excluded from color c, all of whose edges except at most one are already known to have color c. If all edges are known, it reports the path clause as a conflict; if exactly one is unknown, it sets that edge to "not c" with the path clause as the reason. When the solver reaches a complete assignment, the propagator searches it exhaustively for a monochromatic P_m and, if it finds one, rejects the assignment with that path clause. Every clause handed to the solver is a path clause and is written to a log.

The refuted formula is therefore the base formula plus the logged path clauses (plus a cube's unit clauses), a subset of the formula with *all* path clauses. A refutation of the subset refutes the full formula; nothing about the completeness of the propagator is needed. Conversely, a satisfying assignment is accepted only after the exhaustive search, and every coloring the method reports is rechecked by a separate program.

Appendix D The ladder

The rung "P_m on K_n, fixing P_k" is the formula of Section C with the fixed path on $0, \ldots, k-1$, together with all path clauses for P_m. If it is unsatisfiable and every 3-coloring of K_n contains a monochromatic P_k, then by Lemmas C.1 and C.2 every 3-coloring of K_n contains a monochromatic P_m. Every 3-coloring of a larger complete graph then does too, by restriction.

Table 1 lists the rungs. The first fixes nothing; each later rung fixes the path guaranteed by the rung before it. A DRAT proof is a clause-addition and deletion refutation, converted to the line-indexed LRAT format for checking. Each rung up to P_9 on K_{17} is solved as one formula, with a DRAT proof. The rungs establish the upper bounds $R_3(P_m) \leq n$ for $m = 3, \ldots, 9$; the top rung, P_{10} on K_{18}, uses $R_3(P_9) \leq 17$ as established by the rung below it. The earlier P_9 bound is Theorem 8 of Dybizbański, Dzido and Radziszowski [8, arXiv:1207.5851v3, Section 4].

Rung	fixed path	variables	base clauses	logged path clauses
P_3 on K_5	none	110	828	11
P_4 on K_6	P_3	92	527	31
P_5 on K_9	P_4	308	2,135	185
P_6 on K_{10}	P_5	368	2,502	1,683
P_7 on K_{13}	P_6	828	6,234	10,094
P_8 on K_{14}	P_7	928	6,897	60,734
P_9 on K_{17}	P_8	1,720	13,633	520,545
P_{10} on K_{18}	P_9	1,868	14,672	(per cube)

Table 1: The ladder. Each rung from P_3 to P_9 is refuted as a whole; the largest takes a few CPU-minutes. The K_{18} rung is split into cubes (Section E).

The same code also finds colorings where they exist. For example it finds good colorings for P_5 on K_8, P_6 on K_9, P_7 on K_{12} and P_8 on K_{13}, and, with no fixed path, a coloring of K_{16} with no monochromatic P_9; each was confirmed by a separate exhaustive path search.

Appendix E Cube and conquer on K_{18}

This section constructs the split, proves its coverage and joins the refutations.

E.1 The cubes

A cube is a conjunction of literals fixing the colors of some edges. Unit propagation forces the last unassigned literal of a clause when all its other literals are false. The cubes were produced by the solver itself, with the propagator attached. Walking the edges in edge order, at each edge it assumes each color in turn and runs unit propagation (including the path propagator); colors leading to a conflict are dropped, an edge with a single surviving color is fixed without counting as a split, and the walk stops after 10 edges with at least two surviving colors. This produced 15,316 cubes of 11 to 20 literals, and 18 path clauses learned during generation.

Sixteen of these cubes, each with 11 literals, are listed in `data/k18/split/parents.txt`. Each was split once more in the same way, by four further split edges, into 41 children, 656 in all (144 more path clauses were learned during this split). The final list consists of the 15,300 remaining first-level cubes and the 656 children: 15,956 cubes.

E.2 The reduction chain

Suppose a 3-coloring X of K_{18} has no monochromatic P_{10}. Since $R_3(P_9) \le 17$ (the top rung of the ladder, itself resting on the rungs below it, down to P_3 on K_5, which fixes nothing), X contains a monochromatic P_9. By Lemma C.1 a vertex permutation and a color permutation carry X to a good coloring with color 0 on the path $0, 1, \ldots, 8$, and by Lemma C.2 its orbit under the generators contains a good coloring X^* that keeps the fixed path and satisfies every lex-leader inequality. With the auxiliaries set to their meaning, X^* satisfies the base formula and every path clause. By the cover certificate, X^* satisfies some cube of the final list (otherwise it would satisfy the cover formula), and that cube's formula is unsatisfiable by its own certificate: a contradiction. In Lean this whole chain is the theorem `K18_P10_of_certs`, which takes an arbitrary coloring of K_{18} and the certified facts and returns a monochromatic P_{10} (Section F).

E.3 Why the cubes cover every case

Cubes whose propagation fails are dropped during generation, so the list is not complete by construction. Its completeness is proved instead by one more certificate: the *cover formula*, which consists of the base formula, the 162 path clauses learned during generation, and the negation of every final cube (one clause per cube), 30,790 clauses in all, is unsatisfiable.

Lemma E.1. *Let F be a formula, $\mathcal{C}$ a list of cubes, and $\mathcal{P}$ the set of all path clauses. If $F \cup \mathcal{P} \cup \{\neg c : c \in \mathcal{C}\}$ is unsatisfiable and $F \cup \mathcal{P} \cup c$ is unsatisfiable for every $c \in \mathcal{C}$, then $F \cup \mathcal{P}$ is unsatisfiable.*

Proof. A model of $F \cup \mathcal{P}$ satisfies some cube $c \in \mathcal{C}$, since otherwise it falsifies a literal of every cube and so satisfies every $\neg c$. It is then a model of $F \cup \mathcal{P} \cup c$. □

With F the K_{18} base formula, Lemma E.1 and the 15,957 certificates give the unsatisfiability of the top rung, and the ladder then gives $R_3(P_{10}) \le 18$. In Lean this is the theorem combining the cover certificate with the per-cube certificates (`unsat_of_cubes`); the per-cube facts enter through `cubes_of_range`, which turns a theorem `K18.all : ∀ i, 0 ≤ i → i < 15956 → CubeOK i` over the list's indices into the statement for every member of the list, given that the list has length 15,956.

Appendix F The Lean formalization

This section states the formal claim and explains its hypotheses and trusted boundary.

F.1 The statement

The definitions use Mathlib's unordered pairs `Sym2`; a coloring of K_n assigns a color to every pair, and the diagonal pairs are never used.

```
abbrev Col (n : ℕ) := Sym2 (Fin n) → Fin 3

def HasMonoPath (n m : ℕ) (X : Col n) : Prop :=
  ∃ (c : Fin 3) (p : Fin m → Fin n), Function.Injective p ∧
    ∀ (i : ℕ) (h : i + 1 < m), X s(p ⟨i, by omega⟩, p ⟨i + 1, h⟩) = c

noncomputable def ramsey3Path (m : ℕ) : ℕ :=
  sInf {n | ∀ X : Col n, HasMonoPath n m X}
```

```
theorem K18_has_mono_P10 (X : Sym2 (Fin 18) → Fin 3) :
    ∃ (c : Fin 3) (p : Fin 10 → Fin 18), Function.Injective p ∧
      ∀ (i : ℕ) (h : i + 1 < 10), X s(p ⟨i, by omega⟩, p ⟨i + 1, h⟩) = c

theorem K17_no_mono_P10 : ¬ HasMonoPath 17 10 k17Col

theorem R3_P10_eq_18 : ramsey3Path 10 = 18
```

A path is an injective map p from $\{0, \ldots, m-1\}$ to the vertices whose consecutive pairs all have color c, which is the usual (not necessarily induced) path on m vertices. `ramsey3Path m` is the least n for which every coloring of K_n has a monochromatic P_m; the proof of `R3_P10_eq_18` shows that 18 belongs to that set and that no smaller n does, since a good coloring of K_{17} restricts to one of any smaller complete graph.

F.2 What is proved

Lean defines a closed-form encoder `encode n k`, which produces the formula of Section C, and proves for all n and k that it is sound for the colorings: a coloring with the fixed path that satisfies every generator inequality, with the auxiliaries set to their meaning, satisfies every clause (`encode_sat`); a good coloring satisfies every path clause (`paths_ok`). It proves Lemma C.1 (`relabel`), that every generator preserves the fixed path and goodness (`gens_fix`, `act_noMono`), Lemma C.2 (`orbit_min`), the rung argument (`rung`), Lemma E.1 (`unsat_of_cubes`), the chain of rungs, and the lower bound (Section B). The facts about formulas are stated with the DIMACS semantics: a literal is a nonzero integer, an assignment is a predicate on variables, and

```
UnsatWithPaths n m F
```

says that F together with *every* path clause for P_m in K_n has no model.

The proof of `K18_has_mono_P10` uses nine hypotheses of this kind: one for each of the seven rungs, one for the cover formula `encode 18 9 ++ finalCubes.map negCube`, and one for every cube c of `finalCubes`, `encode 18 9 ++ cubeUnits c` (one universally quantified hypothesis over the list, so 15,964 instances in all). The list `finalCubes` is parsed in Lean from the accompanying cube lists, and the cover formula and the cube hypothesis use this same list. The conditional theorems `K18_P10_of_certs` and `R3_P10_eq_18_of_certs` take the nine facts as hypotheses and depend only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound`, as does `K17_no_mono_P10`.

F.3 How the certificates enter

Each instance is proved by a Boolean check `checkUnsat n m F paths proof` and its soundness theorem

```
checkUnsat_sound : checkUnsat n m F paths proof = true → UnsatWithPaths n m F,
```

proved in Lean. The check confirms that every literal of F is nonzero; that each entry of `paths` (a color and m distinct vertices below n) yields a path clause, which Lean rebuilds from the path itself; and that Lean's verified LRAT checker (`Std.Tactic.BVDecide.LRAT`, with its soundness theorem) accepts `proof` as a refutation of F together with those path clauses. The bridge between the two semantics is a proved lemma (`toStd_eval`): under the nonzero-literal condition, an assignment satisfying the formula in the DIMACS semantics above gives an assignment satisfying its translation into the checker's clause type, so the checker's soundness theorem yields `UnsatWithPaths n m F`,

the same predicate that the rung and cover arguments consume. Only the path clauses that the trimmed proof uses are included. Nothing outside Lean decides what counts as a path clause, and the formula checked is a Lean term, not a file: the base formula used in the cube checks is parsed from the accompanying K_{18} formula and proved equal to `encode 18 9` by a comparison that is itself natively evaluated.

Each certificate is stored as a compact string (an LRAT proof with delta-coded hints and implicit clause identifiers). It is decoded by total functions written in Lean. The decoder is outside the logical trusted base: whatever it produces, acceptance implies unsatisfiability by `checkUnsat_sound`. The check is a closed Boolean term, and the kernel accepts its value through `Lean.ofReduceBool`, which evaluates it with Lean's compiled code. The 15,956 cube facts are checked in 162 modules, each covering a contiguous range of indices; a final module joins the ranges into `K18.all`.

F.4 Axioms and trusted base

The axiom report of both main theorems is

```
'R3P10.K18_has_mono_P10' depends on axioms:
  [propext, Classical.choice, Lean.ofReduceBool, Lean.trustCompiler, Quot.sound]
'R3P10.R3_P10_eq_18' depends on axioms:
  [propext, Classical.choice, Lean.ofReduceBool, Lean.trustCompiler, Quot.sound]
```

The two non-standard axioms come from the natively evaluated certificate checks (Lean v4.32.2, pinned in the accompanying project). The trusted base therefore has three parts: Lean's kernel, which checks all the mathematics, including the lower bound and the reduction from colorings to formulas; Lean's compiler and interpreter, whose evaluation of the closed Boolean certificate checks the kernel accepts without reducing them itself (`Lean.ofReduceBool`, `Lean.trustCompiler`); and the compiled modules a build imports, which Lean does not recheck. The evaluated code is the verified LRAT checker, the decoder and the clause and path builders; by `checkUnsat_sound`, only the faithful execution of this code is trusted, not its correctness. Lean does not recheck imported compiled modules, so a reader who wants the whole proof checked on one machine rebuilds the 162 certificate modules (see `checks/checking-record.tex`). The SAT solver, the proof converter, the external checkers and the scripts that produced the certificates are not part of the trusted base of the Lean theorem.

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